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Short-term glucose prediction in Type 1 Diabetes

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Abstract

The use of short-term prediction techniques with the purpose of anticipating hyper- or hypoglycemic episodes for patients with type 1 diabetes has gained a lot of interest in the past years. Both hypoglycemia and hyperglycemia have a major impact and considerable effects in the life of a person with diabetes. The prediction of abnormal values of the blood glucose is a vital step in improving the life quality of a diabetic patient. This paper focuses on developing models for predicting blood glucose levels, to warn type 1 diabetes patients about potential hyper- or hypoglycemic events. Our main objective is to help the daily management of people with type 1 diabetes and to improve their long-term health condition. We will compare four models: a regression model, recurrent neural networks, Long Short-Term Memory, and Gated Recurrent Unit.

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Keywords: glucose level; continuous glucose monitoring; time series; predictive models; type 1 diabetes;

1. Introduction

According to [1], in 2021, there were about 8.4 million people with T1D worldwide, with 0.5 million new cases diagnosed. Type 1 Diabetes (T1D), formerly known as insulin-dependent diabetes or juvenile diabetes, is thought to be an autoimmune disease, in which the immune system destroys insulin-making pancreatic beta cells. Thus, the pancreas produces little or no insulin by itself and requires life-long insulin replacements (through multiple daily insulin injections, or using a programmed insulin pump), in order to avoid diabetic ketoacidosis, which is life-threatening [2]. Glucose monitoring with a continuous glucose monitor (CGM) is essential for diabetes management. CGM devices test, record and store interstitial glucose levels every five minutes, throughout the day and night. The glucose values are measured using sensors that are inserted into the subcutaneous tissue and are transmitted to a receiver, for example a smartphone. At that point, the readings can be visualized and analyzed in order to establish trends, to prevent serious glycemic variabilities. The Glycemic variability (GV) [3] represents the fluctuation of glucose levels between highs

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and lows (hyperglycemia and hypoglycemia), and is considered a potential contributor to T1D-related complications, for example cardiovascular events, kidney failure, stroke, and even death.

The prediction of a hyperglycemia (high blood sugar levels) or a hypoglycemia (low blood sugar level) helps a patient take actions to avoid it before it occurs. An accurate prediction model could improve the quality of life of a T1D patient.

Various time-series prediction algorithms presented in the literature can be used for blood glucose level prediction, such as classical linear models, including Autoregressive (AR), Moving average (MA), Autoregressive Moving Average (ARMA), and Autoregressive Integration Moving Average (ARIMA), or based on artificial intelligence and machine learning, such as Long Short-Term Memory (LSTM) models, or neural networks.

In [4] the authors reviewed a series of 19 papers published between 2005 and 2021, that approached hypoglycemia prediction algorithms and made comparison regarding the data used, the technique on which the predictive model is based, and the accuracy of the model. The analyzed papers tackle various prediction models such as AR models, Classification trees, Decision trees, Support Vector Machine (SVM) based models, Linear projection, Kalman filtering, various neural networks based models, and many more.

Classical statistical models provide a high accuracy when the behaviour of the time series data is not complex and we deal with stationarity. In [5], the authors deal with an ARMA model for short-term glucose prediction in T1D patients. In [6] the authors compare three linear modes, namely AR, ARMA, and ARIMA, as well as four nonlinear algorithms with the purpose of accurately forecasting future blood glucose concentrations and hypoglycemic episodes.

The statistical time-series prediction models are able to capture the linear relationships in time series and obtain good results if the data set is small. However, if we deal with large-scale complex nonlinear time series, there are other options that provide better results [7]. Moreover, in the real world, time series behavior is always nonlinear and, thus, using statistical methods does not have acceptable accuracy for time series analysis. Therefore, artificial intelligence (AI)-based models, including machine learning, or deep learning methods, are more suitable for short-term prediction.

Artificial intelligence (AI) is a fast-growing field that can provide innovative solutions for diabetes care. AI refers to the ability of machines or systems to perform tasks that normally require human intelligence, such as learning, reasoning, decision making, and problem solving. AI can help to improve the diagnosis, prediction, prevention, and treatment of diabetes by using data-driven algorithms that can analyze large amounts of data from various sources such as electronic health records (EHRs), CGMs, wearable devices, mobile apps, social media, and online communities. AI can also help to empower people with diabetes to self-manage their condition by providing personalized feedback, guidance, support, and motivation through digital platforms such as chatbots, virtual assistants, or online coaches [8].

Machine learning (ML), as a sub-field of AI, allows intelligent systems to build appropriate models by learning and extracting patterns in data. Machine learning algorithms are very flexible and this makes them the ideal candidate for solving the problem of finding a perfect balance in which a type 1 diabetic patient must live every day. Machine learning models have been developed in [6], [9], [10], and [11].

Deep learning shows superior performance to traditional ML techniques due to the ability to automatically learn features with higher complexity and representations. Deep learning approaches have recently been applied for blood glucose prediction purpose in [5], [12], [13], and [14].

This paper presents an approach based on various algorithms for predicting BLGs in the future. We will investigate the following models: a regression model (ARIMA) that can estimate the optimal insulin dose based on the predicted blood glucose trend and other factors, such as carbohydrate intake, physical activity, and medication, a LSTM model that can predict the blood glucose trend for the next half an hour or more, recurrent neural networks (RNNs), and a model based on Gated Recurrent Unit (GRU).

A lot of the existing methods and studies described in the literature used simulated data, which are different from real patient data. Others use OhioT1DM data set [15] that contains eight weeks worth of data for each of 12 people with T1D. In this paper, we will use real data collected from a T1D patient since 2019. The experiments were carried out using a Dexocom G6 sensor CGM sensor, which reads the blood glucose level every five minutes, around 288 times per day.

This paper is organized as follows: Section 2 presents the glucose prediction models used for the experimental part, Section 3 details the proposed methodology, in Section 4 we discuss the experimental results, and, finally, in Section 5 we highlight the importance of our contribution and suggest some future work.

2. Prediction Models

The use of advanced time series prediction models like ARIMA, RNN, LSTM, and GRU for blood glucose forecasting is fundamentally justified by their unique strengths in handling complex data trends. ARIMA, with its integration of autoregression, differencing, and moving averages, excels in capturing linear relationships and seasonality in time series data, making it a robust choice for stable, less volatile glucose level trends. RNNs, on the other hand, introduce the concept of memory into the modeling, enabling them to understand and remember patterns over time, which is crucial in capturing the dependencies and fluctuations in blood glucose levels. LSTM and GRU, as extensions of RNN, are specifically designed to overcome the limitations of short-term memory in RNNs. They are adept at learning long-term dependencies, making them highly effective for blood glucose predictions where past information significantly influences future levels. These models can capture the intricate and nonlinear relationships in physiological data, making them essential tools for accurate and reliable blood glucose forecasting, essential for effective diabetes management.

2.1. ARIMA model

ARIMA model, traditionally employed for time series forecasting in various domains, has also been applied to healthcare field of, notably for BGL prediction. While contemporary advances have explored deep learning models for this task, traditional methods like ARIMA still have their place.

ARIMA combines three components: autoregression (AR), differencing (I), and moving average (MA). The AR component models the relationship between an observation and a number of its lagged values. The I component makes the time series stationary, thereby removing time-dependent trends, and the MA component incorporates the dependency between an observation and a residual error from a moving average model applied to lagged observations. This combination renders ARIMA particularly suitable for datasets displaying a clear trend or seasonality.

The ARIMA model, with its inherent simplicity and adaptability, offers a promising avenue for blood glucose forecasting. However, the intricate nature of glucose metabolism, influenced by both endogenous and exogenous factors, means that no single model can capture all the nuances. While ARIMA can serve as a foundational model, especially in scenarios with limited data or computational resources, a holistic approach might necessitate integrating ARIMA with other predictive models, or incorporating external variables for a more comprehensive forecast. As with all predictive tools in healthcare, the objective remains not just accurate forecasting, but facilitating improved clinical decision-making and patient outcomes.

2.2. Recurrent Neural Networks (RNN)

Recurrent Neural Networks (RNNs) have emerged as powerful predictive tools in the realm of time series forecasting. Particularly in the healthcare field, they are being increasingly employed for predicting BGLs, a task of paramount importance for diabetic patients. This section ventures into the efficacy, merits, and constraints of employing RNNs for blood glucose prediction.

RNNs are a subclass of ANNs, specifically designed for sequential data. Unlike traditional feed-forward networks, RNNs possess a memory mechanism that allows them to remember and leverage information from previous time steps. This recursive nature makes them naturally suited to tasks like time series prediction, where temporal dependencies play a crucial role.

The advent of RNNs has undoubtedly bolstered the capabilities of time series forecasting, offering a robust solution to the intricate task of BGL prediction. Their ability to model complex temporal dependencies and non-linearities stands them in good stead. However, the very complexity that empowers RNNs also introduces challenges in terms of computational demands and potential over-fitting.

In the ever-evolving landscape of diabetic care, while RNNs promise a more data-driven and personalized approach, it is imperative to be aware of their limitations. A hybrid approach, possibly combining the strengths of traditional models like ARIMA with RNNs or leveraging other deep learning architectures like LSTMs (which mitigate some of RNNs' shortcomings), might be the path forward. As technology and understanding advance, the primary goal remains steadfast: enhancing the quality of life and health outcomes for diabetic patients.

2.3. Long short-time memory (LSTM)

LSTM networks are a type of RNNs that can learn long-term dependencies in sequential data, such as time series, speech, and text. LSTM networks have been widely used for various applications. LSTM networks can be a suitable method for BGL prediction, as they can capture the temporal dynamics and nonlinear relationships of blood glucose and its influencing factors.

LSTM networks can achieve high accuracy and reliability for different prediction horizons, ranging from 15 minutes to 120 minutes, which is important for diabetes management. However, the prediction accuracy and reliability may decrease as the prediction horizon increases, due to the uncertainty and variability of blood glucose dynamics.

LSTM networks require a large amount of high-quality data to train and test the models. However, the data availability and quality may vary depending on the data sources, such as CGM devices, insulin pumps, food logs, or mobile platforms. Therefore, data preprocessing and feature engineering are essential steps to ensure the validity and reliability of the data.

2.4. Gated Recurrent Unit (GRU) network

The temporal dimension of BGL measurements necessitates the use of time series forecasting techniques that adeptly capture sequential dependencies. GRUs, a recent addition to the RNN family, have garnered attention for their aptitude in handling long-term dependencies, with reduced computational overheads. Introduced as a streamlined alternative to LSTM units, GRUs simplify the recurrent cell's architecture, while retaining its ability to capture long-range dependencies. By consolidating the forget and input gates into a single "update gate", GRUs offer a balance between capturing new information and retaining older memories.

The GRU's advent marks a noteworthy stride in the evolution of RNNs, offering an efficient yet robust alternative to the traditionally used LSTM for time series forecasting. In the domain of BGL prediction, where the interplay of past events profoundly influences future measurements, the GRU's capacity to discern and retain these temporal relationships makes it an asset.

Yet, while the efficiency and reduced computational burdens of GRUs are commendable, it is vital to recognize their potential shortcomings, especially when the complexity of the data warrants a more intricate architecture. As the confluence of deep learning and medical diagnostics continues to deepen, the judicious selection, adaptation, and optimization of these tools become paramount in driving clinically meaningful predictions and interventions.

3. Methodology

Data collection and storing: the CGM device measures the BGL every few minutes and sends the data to a cloud server. The server stores the data in time series, which can be queried by the workflow. The workflow can ingest historical data from the time series database, or use real-time data from the CGM device. The data is formatted as a sequence of glucose values with timestamps. For our experiments we used local stored data.

Data preprocessing: the preprocessing steps include: data augmentation (if only 1 value is missing, we add a value using the average of the neighbor values), data smoothing (the data is smoothed using a moving average filter to reduce the noise and outliers in the data), data normalization - the data is normalized using Min-max normalization, and data splitting (the data is split into sequences the same size of the input data, then into subsets for training, and testing).

The **prediction models** selected for this paper are: ARIMA, RNN, LTSM, and GRU. The primary challenge lay in determining the optimal order for the ARIMA model. The order of an ARIMA model is denoted by (p, d, q), meaning the autoregressive, differencing, and moving average terms, respectively. To ensure a comprehensive search, our experiments spanned orders from 0 to 5 for each of the p, d, and q parameters. This exhaustive search was paramount, as the performance of ARIMA models can be highly sensitive to the chosen order.

Regarding RNN based model, the foundation for our analysis was an RNN with two layers, each comprising 50 neurons, using the Rectified Linear Unit (ReLU) activation function. The model was trained with a batch size of 32 for a span of 20 epoch. The last step consists in testing and evaluation using several evaluation criteria. This preliminary configuration proved promising results, encouraging further exploration. Extending the output size revealed an

incremental increase in errors, which was expected as long-term predictions tend to amplify uncertainties. Interestingly, switching the activation function from ReLU to tanh showed an improved performance, hinting that the tanh function's non-linearity and bounded nature might be more suited for this specific prediction task. Similarly, when the model's architecture was altered, either by removing or adding layers, there was not a significant enhancement, underscoring the necessity to strike a balance between model complexity and overfitting. Furthermore, modifying training parameters like batch size and the number of epochs revealed valuable insights. A larger batch size showed slight improvement in metrics, whereas doubling the training epochs surprisingly led to a considerable increase in error (see Table 2), indicating potential overfitting.

Moving on to LSTM model, we started with a fundamental configuration of a single layer containing 50 neurons and using the ReLU activation function, the focus remained on forecasting with a 12-step output size. The model underwent training for 20 epochs with a batch size of 32. The research was undertaken in phases, wherein based on preceding outcomes, attributes of the LSTM model's architecture or training hyperparameters were adjusted. During our initial experiments, a warning pertaining to the LSTM layer's inability to utilize cuDNN kernels arose. cuDNN (CUDA Deep Neural Network) is a GPU-accelerated library that can significantly enhance the speed of neural network computations. The use of the ReLU activation function prevented the model from meeting the criteria for cuDNN optimization, thereby resorting to generic GPU kernels and incurring computational inefficiencies. To address this and avail the speed benefits of cuDNN, the activation function was switched to tanh. Not only did this accelerate model training, but it also improved the model's performance, evident from the reduction in error metrics. Subsequent adjustments in the model's architecture, such as increasing the number of neurons, displayed further enhancements in prediction accuracy. Training the model for an extended period of 100 epochs with 100 neurons proved to be the most effective configuration, revealing a substantial decrease in all error metrics (see Table 3).

For GRU based model, our initial experiment began with a modest configuration: two layers, each comprising 50 neurons, activated by the ReLU function. The choice of the Adam optimizer and a Mean Squared Error loss function steered our model through 20 epochs with a batch size of 32. While the results showcased promise, we perceived the potential for optimization. The subsequent tweak was a switch to the tanh activation function, which yielded a substantial improvement (shown in Table 4). Encouraged by this leap, the next set of experiments revolved around increasing the network depth. Adding a third layer slightly escalated the MAE and sMAPE but showcased the resilience of GRU to the potential pitfalls of increasing complexity. With a further addition, moving to four layers, the metrics remained fairly consistent, emphasizing the stability of the architecture. However, pushing the network to its limits with 100 neurons in each of its four layers resulted in a slight degradation in performance. It became evident that while depth and width are essential, there's an intricate trade-off to be navigated in the GRU architecture, akin to other neural networks.

4. Experimental Results

4.1. Data set

The data used in this study comes from Dexcom G6 CGM device. It consists of three components: a sensor, a transmitter, and a display device. The sensor is a small, flexible wire that is inserted under the skin of the abdomen or the upper arm. The sensor measures the glucose levels in the interstitial fluid every five minutes and sends the data to the transmitter. The transmitter is a small device that attaches to the sensor and communicates with the display device through a wireless communication. The display device can be a compatible smart device (such as a smartphone or a smartwatch) or a dedicated Dexcom receiver. The display device shows the current glucose level, the trend arrow, and the historical data in easy-to-read graphs. The display device can also alert the user when the glucose level is too high or too low, or when it is changing rapidly. The research is based on data collected from a single individual, one of the authors, ensuring consistency in the data source and eliminating inter-individual variability.

4.2. Evaluation criteria

The evaluation was done using the Mean Squared Error (MSE), the Mean Absolute Error (MAE), the Mean Absolute Percentage Error (MAPE), and the Symmetric Mean Absolute Percentage Error (sMAPE) error metrics. MAE

is a metric used to measure the average absolute difference between predicted and actual values in a dataset, while MSE measures the average of the squared differences between predicted and actual values. MAE is less sensitive to outliers compared to MSE. Both MAE and MSE are useful for evaluating the performance of predictive models, and the choice between them depends on the specific characteristics of the problem and the desired sensitivity to errors. MAPE is one of the most popular metrics for evaluating the prediction performance. A lower MAPE indicates a more accurate forecast. sMAPE is similar to MAPE but has the advantage of being symmetric. Both MAPE and sMAPE are expressed as percentages, with lower values indicating better forecast accuracy.

The formulas for all the metrics are given below:

$$\text{MSE} = \frac{1}{N}\sum_{i=1}^{N}(Y_i - \hat{Y}_i)^2, \tag{1}$$

$$\text{MAE} = \frac{1}{N}\sum_{i=1}^{N}|Y_i - \hat{Y}_i|, \tag{2}$$

$$\text{MAPE} = \frac{1}{N}\sum_{i=1}^{N}\left|\frac{Y_i - \hat{Y}_i}{Y_i}\right| \times 100 \tag{3}$$

$$\text{sMAPE} = \frac{1}{N}\sum_{i=1}^{N}\frac{\left|Y_i - \hat{Y}_i\right|}{\frac{|Y_i|+|\hat{Y}_i|}{2}} \times 100, \tag{4}$$

where N is the length of the time series, Y_i is the glucose concentration measured by the CGM sensor at time i and $\hat{Y}_i$ is the predicted value.

4.3. Results

In our quest to identify the most suitable model for predicting blood glucose levels, we embarked on a series of experiments leveraging the ARIMA modeling technique. Our experiments were primarily structured around forecasting using an input window of 36 values. We explored predictions for various output sizes: 1, 6, and 12. The results fare shown in Table 1.

Table 1. Performance Metrics for ARIMA(2,1,0)

Configuration	MSE	MAE	MAPE (%)	sMAPE (%)
1-output size configuration	1.4167	0.9022	0.6254	0.6262
6-output size configuration	52.3313	4.3630	3.0610	3.1035
12-output size configuration	388.8692	11.1705	7.6934	7.9466

For the RNN network, we tried multiple experiments modifying various hyperparameters to observe how the desired metrics are influenced. The base architecture for RNN was using ReLU, activating 2 layers of 50 neurons. The results are shown in Table 2.

Table 2. Performance Metrics for RNN Configurations

Configuration	MSE	MAE	MAPE (%)	sMAPE (%)
Output size 1	1.6360	0.9660	0.6865	0.6871
Output bumped to 6	34.5532	3.9494	2.7022	2.7064
Output bumped to 12	360.5807	12.3785	8.6932	8.2952
Activation switched to tanh	257.2285	11.0432	7.6346	7.3645
Minimized MAE	265.7766	9.5134	6.3934	6.4686
One layer removed	264.6086	10.3135	7.4424	7.1421
Additional layer added	273.3843	10.2006	6.8830	6.8718
Batch size of 64	254.8469	10.2463	6.8216	6.9196
Trained for 40 epochs	695.2883	18.4869	13.1249	12.3488
100 neurons per layer	218.0462	9.5605	6.512	6.4178

The next experiment was centered around the LSTM model. The result are shown in the table below.

Table 3. Performance Metrics for LSTM Configurations

Configuration	MSE	MAE	MAPE (%)	sMAPE (%)
Initial (ReLU, 1 layer, 50 neurons)	452.3543	15.5132	11.3514	10.6158
Activation switched to tanh	292.5748	11.2585	7.3335	7.4949
100 neurons per layer	239.0717	10.1737	7.7975	7.5406
100 neurons, trained for 100 epochs	134.0968	7.2285	4.7631	4.8343

The last experiment was using GRU units for a model. The results are shown in Table 4:

Table 4. Performance Metrics for GRU Configurations

Configuration	MSE	MAE	MAPE (%)	sMAPE (%)
2 layers, ReLU, 50 neurons	574.141	16.785	13.863	12.539
2 layers, tanh, 50 neurons	217.583	9.649	6.650	6.491
3 layers, tanh, 50 neurons	216.957	10.306	7.455	7.505
4 layers, tanh, 50 neurons	217.792	9.683	6.475	6.614
4 layers, tanh, 100 neurons	249.243	10.018	7.065	6.857
4 layers, tanh, 100 neurons, 100 epochs	274.052	10.277	7.167	7.043

4.4. Study findings

The ARIMA model's performance, especially with the 1-output size, is commendable. While the inability to run on GPU can be a limitation in terms of scalability and speed, the model's capability to handle non-normalized data and its demonstrated accuracy suggests it remains a viable option for certain prediction tasks. However, as with any model, it's essential to weigh the benefits against the constraints in the context of the specific application and available resources.

Exploring standard RNNs offered an enlightening perspective on the intricacies of model design, the choice of hyperparameters, and the trade-offs involved. While certain configurations showed potential, it's evident that iterative model refinement and robust validation are crucial to harnessing the full predictive power of RNNs for such intricate tasks

In time-series forecasting, the decision between GRU and LSTM often emerges as a point of contention. Both architectures offer their unique strengths. LSTMs, with their three-gate structure, possess an unparalleled ability to model long-term dependencies in sequences. However, this prowess comes with the baggage of computational overhead and a steeper learning curve. GRUs, on the other hand, with their more streamlined structure, present a more efficient alternative. In our experiments, we found the GRU to be highly competitive, often outmatching or rivaling the LSTM in performance metrics, but with fewer parameters and often faster convergence.

The emergence of GRU as a formidable force in the world of RNN has been substantiated by our experiments. Its capability to balance efficiency with performance makes it an attractive choice for time series forecasting, especially in scenarios where computational resources are at a premium. While the journey of hyperparameter and architectural optimization is laden with challenges, the rewards, as seen through our results, are worth the pursuit.

The preliminary experiments that measure the performance of the model when running on data from other patients shows promising results.

5. Conclusions

Our venture into various predictive models, from ARIMA and RNNs to LSTMs and GRUs, marked a significant phase. These models, each with its strengths and idiosyncrasies, were the analytical engines that powered our glucose prediction. Rigorous experiments and relentless optimisation were carried out, ensuring the highest levels of accuracy and reliability. Our findings highlighted the GRU as a particularly potent tool, balancing computational efficiency with predictive prowess to close the gap between GRU and LSTM.

The next step in our research journey is to compare the results of the models with data from different patients with similar systems. The next goal is to use the data from known public data sets.

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